

E-Commerce Sales Analytics Dashboard Project Report

Project Overview

This project focuses on analyzing an e-commerce sales dataset to discover meaningful business insights, evaluate customer purchasing patterns, and build a machine learning model capable of predicting future sales. The project combines data cleaning, exploratory data analysis (EDA), machine learning, and dashboard development to support data-driven business decision-making.

Objectives

- Import and clean the e-commerce sales dataset.
 - Perform exploratory data analysis to identify sales trends.
 - Analyze customer purchasing behavior and regional performance.
 - Build multiple machine learning models for sales prediction.
 - Compare model performance and select the most accurate model.
 - Develop an interactive dashboard for business users.
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Dataset Description

The project uses the **Sample Superstore** dataset from Kaggle, containing sales transactions across different regions, customer segments, categories, and products.

Main attributes include:

- Order Date
- Ship Date
- Customer ID
- Customer Name
- Segment
- Region
- State
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

Data Preprocessing

The following preprocessing steps were performed:

- Removed duplicate records.
- Checked and handled missing values.
- Converted date columns into datetime format.
- Created new features including Year, Month, Quarter, Day Name, and Profit Margin.
- Exported a cleaned dataset for machine learning.

Exploratory Data Analysis

Several visualizations were created to understand business performance, including:

- Monthly Sales Trend
- Category-wise Sales
- Top Selling Products
- Regional Sales Performance
- Customer Segment Analysis
- Top Customers
- Profit Analysis
- Discount Impact
- Correlation Heatmap
- Sales and Profit Distribution

Machine Learning Models

Three regression models were trained and evaluated:

1. Linear Regression
2. Decision Tree Regressor
3. Random Forest Regressor

The models were compared using:

- Mean Absolute Error (MAE)
 - Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score
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Model Comparison

After comparing all evaluation metrics, the **Decision Tree Regressor** achieved the best overall performance for this dataset.

Reasons for selecting Decision Tree:

- Highest prediction accuracy among the tested models.
- Better handling of non-linear relationships.
- Lower prediction error compared with the other models.
- Easy interpretation of decision-making patterns.
- Fast prediction and suitable for business applications.

Therefore, the Decision Tree Regressor was selected as the final prediction model for deployment.

Dashboard Development

An interactive Streamlit dashboard was developed featuring:

- Revenue KPI
- Profit KPI
- Order KPI
- Monthly Sales Trend
- Category-wise Analysis
- Regional Performance
- Top Selling Products
- Customer Insights
- Sales Prediction

The dashboard enables users to interactively explore business performance and generate sales predictions.

Business Insights

The analysis revealed several important findings:

- Sales varied significantly across different months.
- A small number of products generated a large percentage of total revenue.
- Certain categories consistently outperformed others.
- Regional sales performance differed considerably.
- Repeat customers contributed substantially to overall sales.
- Discount strategies had a noticeable impact on profitability.
- High-profit products should receive increased marketing attention.

- Seasonal sales patterns can be used for demand forecasting.
 - Customer segmentation supports targeted marketing campaigns.
 - Data-driven inventory planning can reduce operational costs.
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Conclusion

This project successfully transformed raw sales data into actionable business intelligence through comprehensive data analysis, machine learning, and dashboard visualization. Among the evaluated algorithms, the **Decision Tree Regressor** delivered the best predictive performance and was selected as the final model for sales prediction.

The completed solution provides an effective decision-support system that enables organizations to monitor sales performance, identify growth opportunities, and forecast future sales with improved accuracy. The integration of data analytics, predictive modeling, and interactive visualization demonstrates the practical application of data science techniques in solving real-world business problems.